

**MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS**  
**NANYANG TECHNOLOGICAL UNIVERSITY**

Tutorial 8

21/10/2021

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We will discuss several questions from Chapter 6 and Chapter 7 in Ross' book.

**Tutorial Questions**

- (1) The joint density of  $X$  and  $Y$  is given by

$$f(x, y) = \begin{cases} xe^{-(x+y)}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Are  $X$  and  $Y$  independent? If, instead,  $f(x, y)$  were given by

$$f(x, y) = \begin{cases} 2, & 0 < x < y, 0 < y < 1; \\ 0, & \text{otherwise,} \end{cases}$$

would  $X$  and  $Y$  be independent?

- (2) Suppose that  $X, Y, Z$  are independent random variables that are each equally likely to be either 1 or 2. Find the probability mass function of

- (a)  $XYZ$ ;
- (b)  $XY + YZ + ZX$ ;
- (c)  $X^2 + YZ$ .

- (3) Consider two components and three types of shocks. A type 1 shock causes component 1 to fail, a type 2 shock causes component 2 to fail, and a type 3 shock causes both components 1 and 2 to fail. The times until shocks 1, 2, and 3 occur are independent exponential random variables with respective rates  $\lambda_1$ ,  $\lambda_2$ , and  $\lambda_3$ . Let  $X_i$  denote the time at which component  $i$  fails,  $i = 1, 2$ . Find  $P\{X_1 > s, X_2 > t\}$ .

*The random variables  $X_1, X_2$  above are said to have a joint bivariate exponential distribution.*

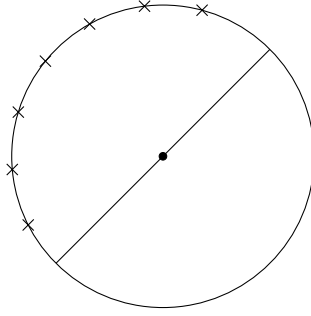
- (4) Let

$$f(x, y) = 24xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad 0 \leq x + y \leq 1$$

and let it be 0 otherwise.

- (a) Show that  $f(x, y)$  is a joint density function.
- (b) Find  $\mathbb{E}(X)$  and  $\mathbb{E}(Y)$ .
- (c) Show that  $X$  and  $Y$  are not independent.

- (5) Suppose that  $n$  points (assuming that  $n > 2$ ) are independently chosen at random on the circumference of a circle, and we want the probability that they all lie in some semicircle. That is, we want the probability that there is a line passing through the center of the circle such that all the points are on one side of that line, as shown in the following diagram:



Let  $P_1, \dots, P_n$  denote the  $n$  points. Let  $A$  denote the event that all the points are contained in some semicircle, and let  $A_i$  be the event that all the points lie in the semicircle beginning at the point  $P_i$  and going clockwise for  $180^\circ$ ,  $i = 1, \dots, n$ .

- (a) Express  $A$  in terms of the  $A_i$ .
- (b) Are the  $A_i$  mutually exclusive?
- (c) Find  $P(A)$ .

- (6) The joint density function of  $X$  and  $Y$  is given by

$$f(x, y) = xe^{-x(y+1)}, \quad x > 0, y > 0$$

- (a) Find the conditional density of  $X$ , given  $Y = y$ , and that of  $Y$ , given  $X = x$ .
- (b) Find the density function of  $Z = XY$ .

- (7) Let  $X_1, \dots, X_n$  be independent and identically distributed random variables having CDF  $F$  and density  $f$ . We sort them in increasing order  $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ , where each  $X_{(i)}$  is a random variable. The quantity  $M := [X_{(1)} + X_{(n)}]/2$ , defined to be the average of the smallest and largest values in  $X_1, \dots, X_n$ , is called the midrange of the sequence. Show that its CDF is

$$F_M(x) = n \int_{-\infty}^x [F(2x - t) - F(t)]^{n-1} f(t) dt.$$

## Assignment Questions

- (1) Let  $X, Y, Z$  be independent and uniformly distributed over  $(0, 1)$ . Compute  $P\{X \geq YZ\}$ .

- (2) The joint density of  $X$  and  $Y$  is

$$f(x, y) = c(x^2 - y^2)e^{-x}, \quad 0 \leq x < \infty, \quad -x \leq y \leq x.$$

Find the conditional CDF of  $Y$ , given  $X = x$ .

- (3) Two points are selected randomly on a line of length  $L$  so as to be on opposite sides of the midpoint of the line. [In other words, the two points  $X$  and  $Y$  are independent random variables such that  $X$  is uniformly distributed over  $(0, L/2)$  and  $Y$  is uniformly distributed over  $(L/2, L)$ .] Find the probability that the distance between the two points is greater than  $L/3$ .
- (4) A complex machine is able to operate effectively as long as at least 3 of its 5 motors are functioning. If each motor independently functions for a random amount of time with density function  $f(x) = xe^{-x}$ ,  $x > 0$ , compute the density function of the length of time that the machine functions.
- (5) An accident occurs at a point  $X$  that is uniformly distributed on a road of length  $L$ . At the time of the accident, an ambulance is at a location  $Y$  that is also uniformly distributed on the road. Assume that  $X$  and  $Y$  are independent. Find the expected distance between the ambulance and the point of the accident.
- (6) The county hospital is located at the center of a square whose sides are 3 miles wide. If an accident occurs within this square, then the hospital sends out an ambulance. The road network is rectangular, so the travel distance from the hospital, whose coordinates are  $(0, 0)$ , to the point  $(x, y)$  is  $|x| + |y|$ . If an accident occurs at a point that is uniformly distributed in the square, find the expected travel distance of the ambulance.